Can drug signatures shift muscle-cell context in an exercise-like direction?

A cell-type-aware, in-silico pilot using real reference nuclei and public expression signatures

PUBLIC DATA → PAIRED SIMULATED SHIFTS → DIRECTION + CELL-TYPE COMPARISON

1 Use real biological inputs

GTEx skeletal-muscle nuclei
66 nuclei · 22 annotated cell types
3 nuclei sampled per type

Exercise gene signature
Endurance vs control · 24 h · 16 genes

5 GEO drug signatures
Metformin (2 doses), simvastatin, rosuvastatin and GW501516 / Cardarine

2 Simulate paired shifts

Same reference nucleus

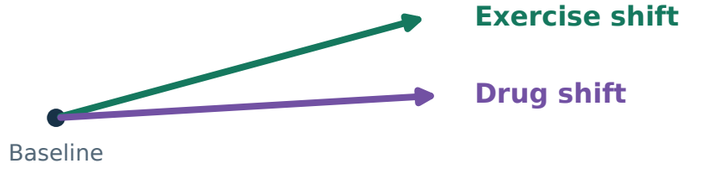
Baseline	Exercise	Drug
Unchanged	Apply exercise shifts	Apply drug shifts

Re-rank the top 300 expressed genes
Gene-rank sentences → C2S embeddings

Bounded heuristic shifts from signed gene scores; not measured post-treatment cells.

3 Compare shift directions

Subtract the same baseline embedding; compare the two resulting shift vectors.



Conceptual geometry, not plotted results

Cosine similarity: Δ_{drug} vs $\Delta_{exercise}$
Average valid paired scores within each cell type

Independent gene-level comparison
Expression-weighted Pearson correlation; requires ≥3 shared, expressed signature genes.

WHAT WE WANTED TO TEST

Direction
Do drug and exercise shifts align or point in opposing directions?

Cell-type specificity
Does alignment differ between myocytes and other muscle-resident cells?

Added value of C2S
Does embedding alignment agree with or add insight beyond gene-level similarity?

WHAT THIS RUN COULD RESOLVE

10/66	45/330	0/330	Limited resolution
exercise sentences changed	finite C2S comparisons	finite gene-level comparisons	Hypothesis-generating only

Positive cosine = alignment; negative = opposition; unresolved ≠ no biology. Neither is a causal exercise-mimetic / blunting claim.

Methods: C2S final-layer mean embeddings · same-nucleus shifts · signed-statistic rank proxy. Counts: saved pilot-run outputs.